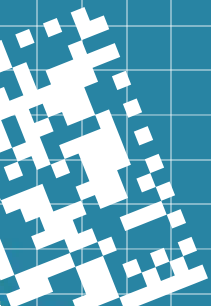
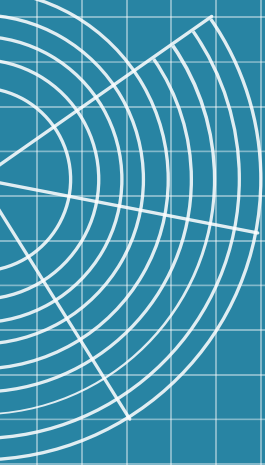
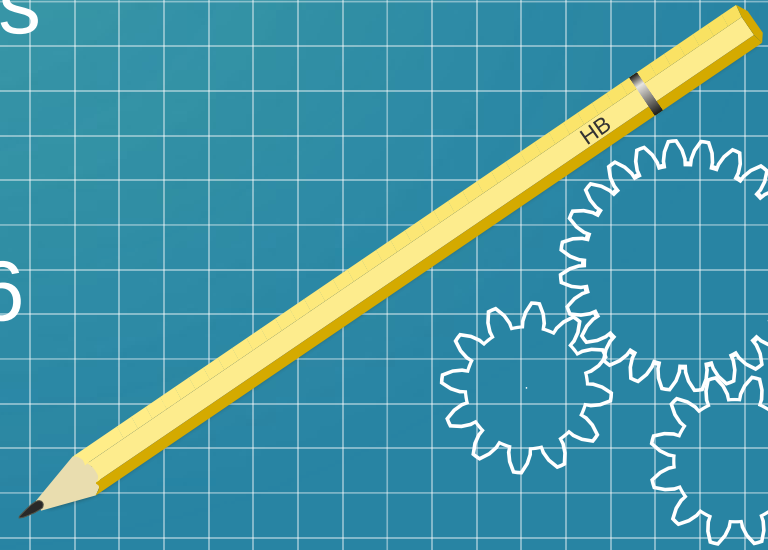
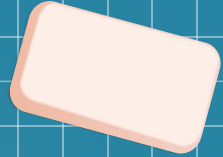


Let's Talk About Digital Accessibility

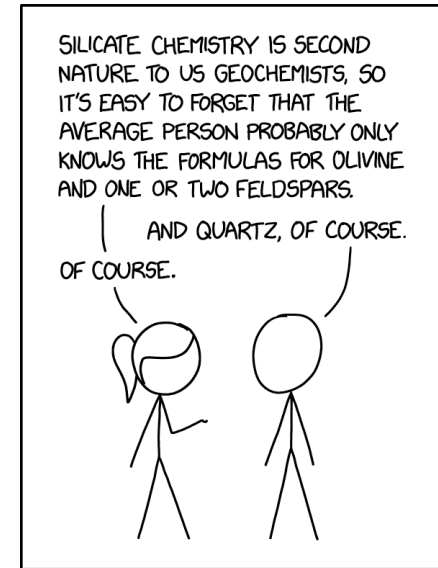
America Masaros

WisMATYC
September 2026



Speaker Background

- I did elementary school homework in Word 1.0 on a Macintosh SE
- I made my first web page in 1996
 - We thought <blink> was pretty cool
 - Alt text used to be for people with slow internet connections
- I was in Harper's pilot group for Blackboard Ally in Fall 2020



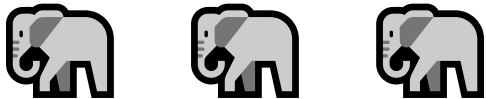
EVEN WHEN THEY'RE TRYING TO COMPENSATE FOR IT, EXPERTS IN ANYTHING WILDLY OVERESTIMATE THE AVERAGE PERSON'S FAMILIARITY WITH THEIR FIELD.

<https://xkcd.com/2501/>

Universal Design for Accessibility



- Accessibility is Good
- Undiagnosed or Undisclosed Disabilities
 - Disclosing Disabilities is a Pain, Actually
- Curb-Cut Effect

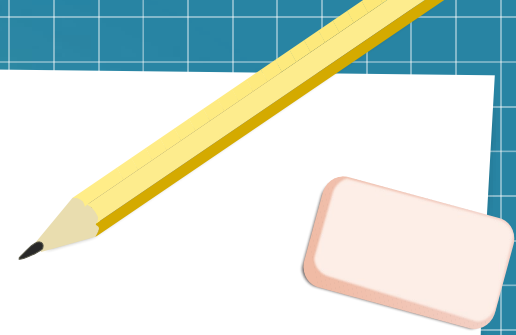


BUT



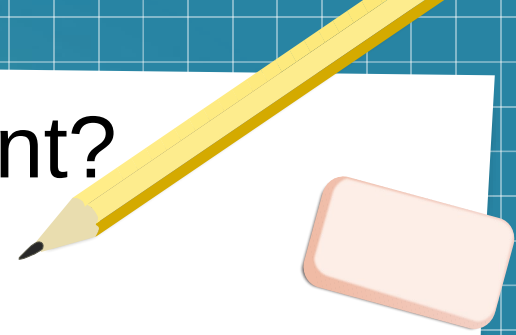
- Competing Access Needs

Accessible Documents



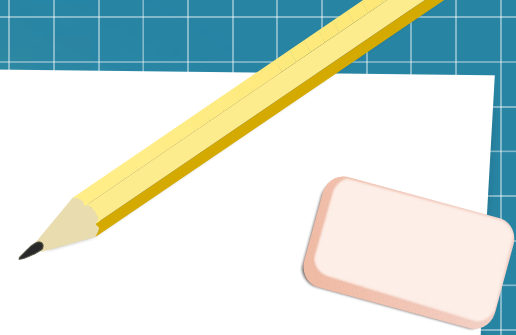
- Three Goals
 - An overall accessible course
 - Accessible documents
 - Documents that software tools **think** are accessible
- These goals can be at odds!

What's an Accessible Document?



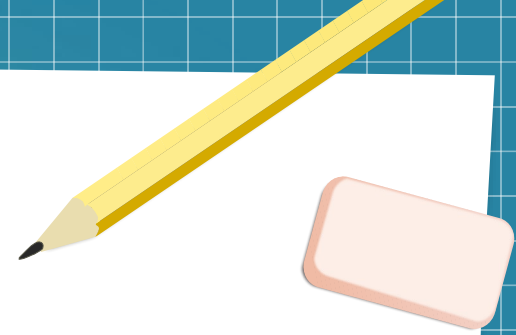
- Screen reader friendly structure
 - Text is actually text
 - Images have “alt text” (to be read when the image would be shown)
 - Tables have headers (to help give them meaning)
- Colorblind friendly
 - Colorblindness is *really common* (esp. in people with one X chromosome) and may not be registered as a “disability.”
 - Don’t encode important information ONLY with color
 - “Red dashed graph vs green solid graph” not “red graph vs green graph”
 - High contrast

About HTML Accessibility



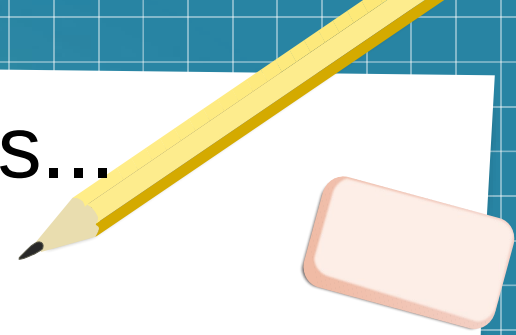
- You should probably learn a little HTML.
 - Khan Academy
 - HTML for People: <https://htmlforpeople.com/>
 - W3Schools.com HTML tutorial:
<https://www.w3schools.com/html/>
- Headings: `<h1>` `<h2>` `<h3>` etc
- Tables: `<thead>` and `<th>`
- Formatting: appearance vs. syntax

PDF Considered Harmful?



- PDF is not a Web format
 - *Web pages* should be XML/HTML
- Documents designed for print should be supplied as PDF
 - Consistent presentation
 - Platform independence
 - Layout conveys information
- Documents intended for on-screen viewing should not
 - More on this later
- PDF is a proprietary format 🙄

Speaking of proprietary formats...

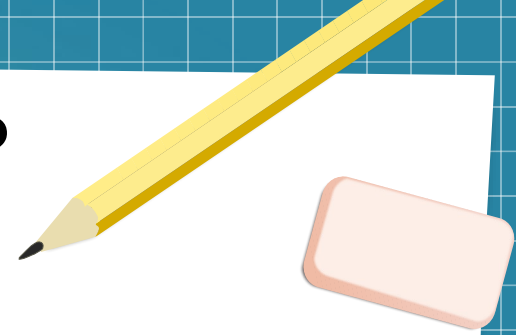


You need Adobe Acrobat Pro

- The last version of Acrobat with a perpetual license was Acrobat Pro 2020. Sales ended in 2024.
- Do with this information what you will.

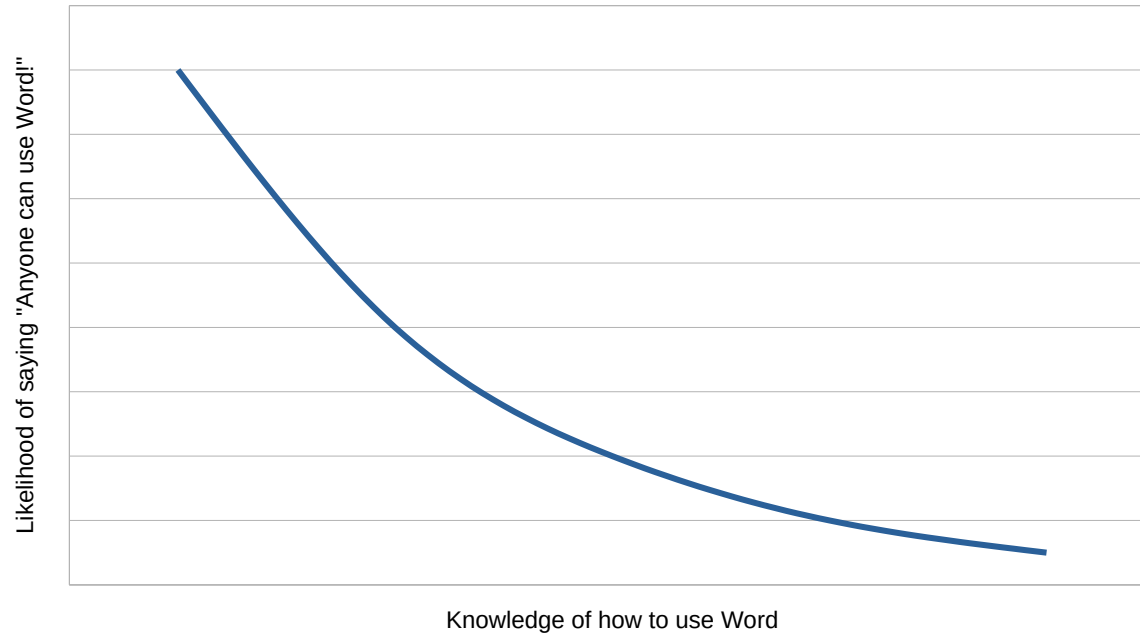
Why Do You Need Acrobat?

- Alternative tools have poor support for “tags”
- Fix scanned documents
- Fix annotated documents
- Fix old PDF files
- **Figure out what the problem actually is**

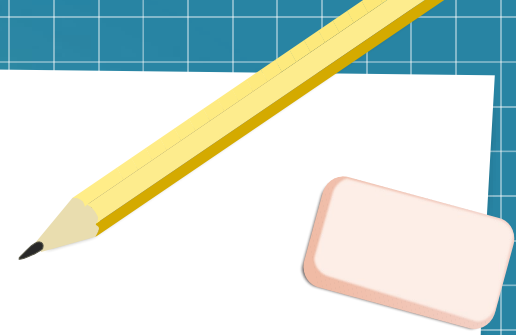


The Good News

- Microsoft Word makes this easy
 - If you actually learn how to use Word

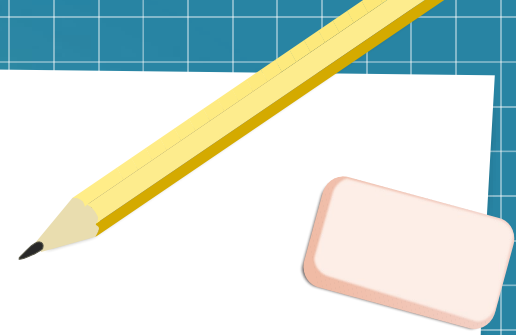


How to Use Word



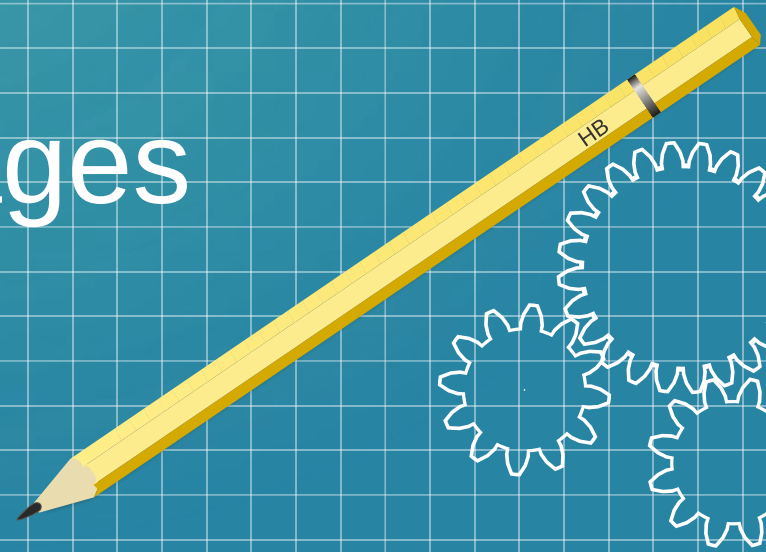
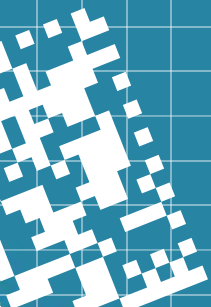
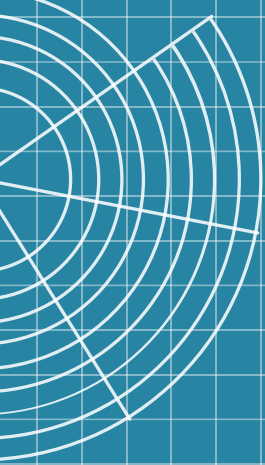
- “Outline” view ➡
 - Your document should make sense from the headers.
- “Table Design” ribbon ➡
 - “Header,” “First Column,” or both
 - Tables are for tables – consider using columns or tabs for layout
- Images ➡
 - In-Line
 - Alt Text
- Check Accessibility ➡
 - Happens automatically when you try to save to PDF!
- “Title” is carefully hidden ➡

Other Word Processors

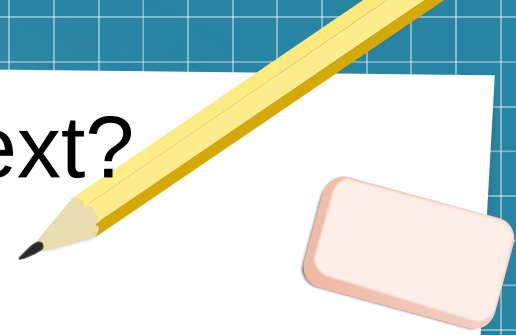


- LibreOffice
 - [Alt][8] or Tools Accessibility Check...
 - Less user-friendly than MS Word
 - Catches more issues (title, line spacing)
 - Produces standards compliant PDFs
 - Does *not* auto-generate alt-text for equations
 - This is both good and bad...
- Google Docs
 - Prompts for image alt text
 - Can't find accessibility checks
 - Auto-generates alt-text for equations (at least, native equations...)
 - Does *not* produce standards-compliant PDFs

Alt Text for Images



Who will “see” (hear) your alt text?



- Lots of people use a screen reader!
- Reading-Based Disabilities (Dyslexia)
- Attention-Based Disabilities (ADHD)
- Multi-Taskers
- Vision-Based Disabilities
 - Visually Impaired
 - Blind (Low Vision)
 - Blind (Sightless)

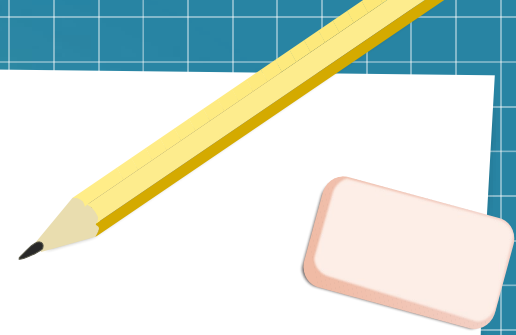
Alt Text for Images – Why Is This Image Here?

A yellow pencil with a pink eraser is positioned diagonally in the top right corner of the slide. The pencil is pointing towards the bottom left, and the eraser is at the top right.

- What information should a sighted person get by looking at the image?
- Common kinds of information in math images:
 - Numbers that can be read from the graph
 - “A line with slope -3 and y-intercept $(0,4)$.”
 - General shape of the graph
 - “A parabola and a line. They intersect at two points, marked as P and Q.”
 - Data also shown elsewhere
 - “A histogram for the preceding data table. It is approximately bell-shaped.”
- Think about what you want to **communicate**.

Mark-Up Languages

- MathML for math formulas
- ChemicalML for molecular diagrams
- Others?
- Can it be added in Word?
- Requires PDF-UA/2.0 support

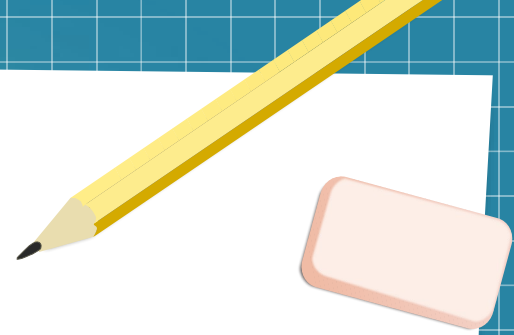


But what if —



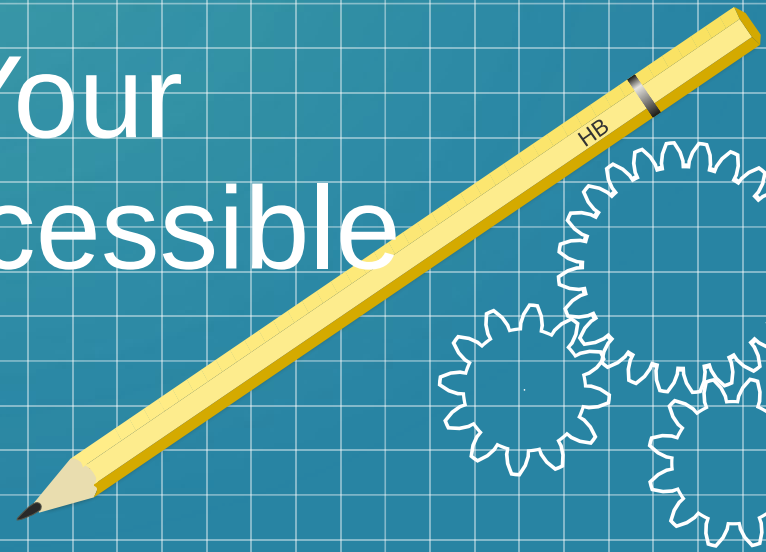
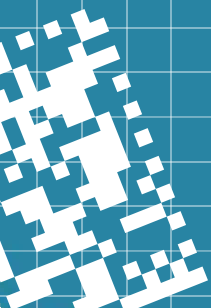
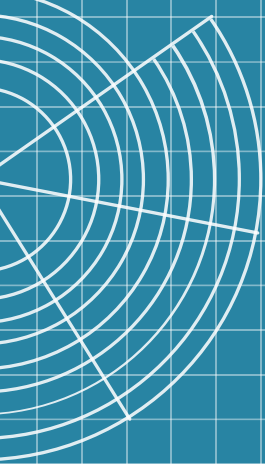
- What if the information is, “This is what this picture looks like!”
 - “Image Described In Text”
 - “A graph of the preceding equation”
 - “Diagram for [purpose]”
- Don’t be afraid to mark it decorative
 - If you have a blind (sightless) student, you will work with disability services to develop tactile alternatives.
 - The TI-84 is a useless device to anyone who can’t see your TI-84 screen shots. Mark them decorative or give short tags.
 - DESMOS is the accessible alternative for blind (sightless) users

Don't Panic!



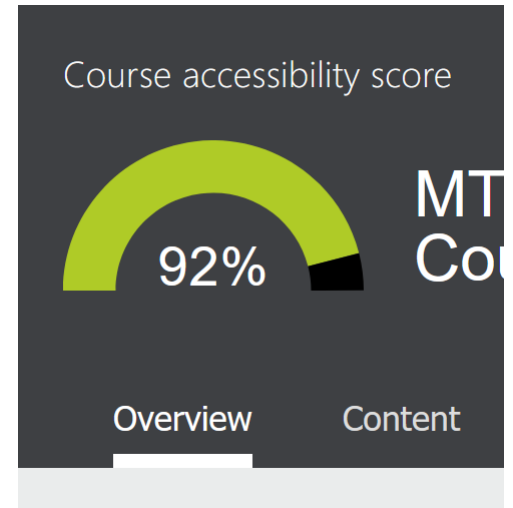
- **Bad alt text is better than no alt text.**
- Short alt text is better than excessively long alt text.
- You are **probably** already describing the image in text.
- You can edit later.

How to Make Your Scanned Notes Accessible



Posting Scanned Notes

- Posting your class notes makes your **course** more accessible!
- A hand-written document will never be an accessible **document**.
- The **information** should be available in your course in an accessible format.
- But, you *can* get your accessibility checker off your back.



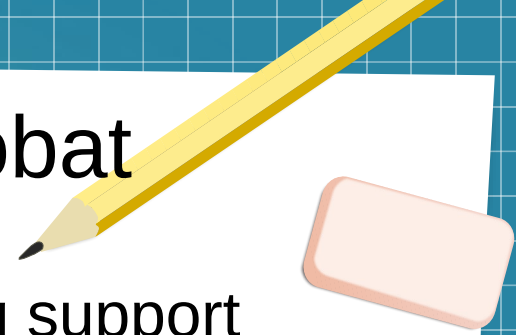
Making Course Notes Accessible



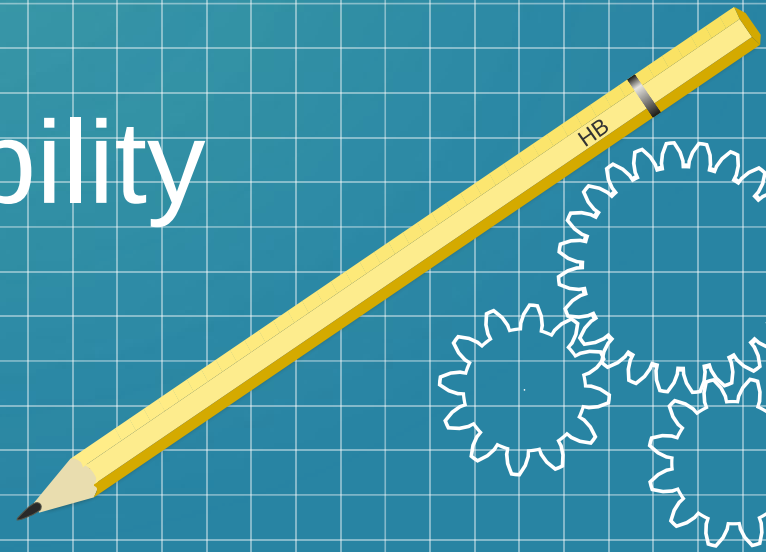
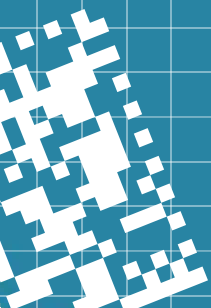
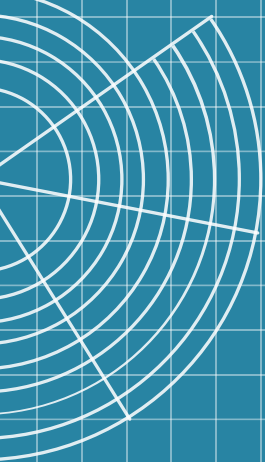
- Ally does not not like “Scanned and OCR’d” notes
- It likes “annotated” (on a computer) notes better
- If you are annotating a PDF, the PDF you start with should be accessible.
- This is where you need Acrobat!
 - Alternative OCR software doesn’t handle tags.
 - Run Acrobat’s “Prepare for Accessibility”
- Balance time vs. quality
 - These are ephemeral documents

Making It Happen Without Acrobat

- Blackboard is working on better OCR and tagging support
- Currently promised for October update – haven't yet tested
- Other LMS – ask your LMS provider (and bother them if they aren't working on it!)

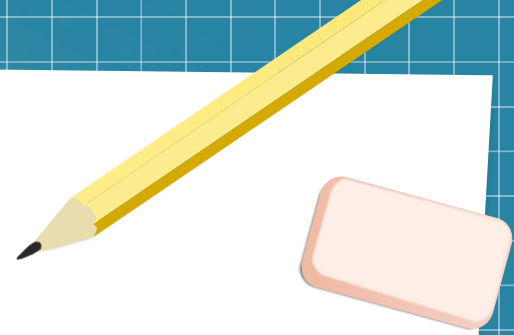


LaTeX Accessibility



The Good News

- Accessible LaTeX output now exists!
- (As of LaTeX 3.0; make sure you're up to date.)
- ... Kind of.



The Bad News



- As of right now, Tagged PDF is incompatible with basically all packages.
 - That's an exaggeration (but not much)
 - TikZ works!
 - If it messes with tables, headings, or lists, it's probably broken
- Most of them quietly break the tagging.
- Some of them just won't run.
- Also you need to run in LuaLaTeX
 - This WILL mess up your fonts and spacing.

The Ugly News—Let's Talk PDF Standards



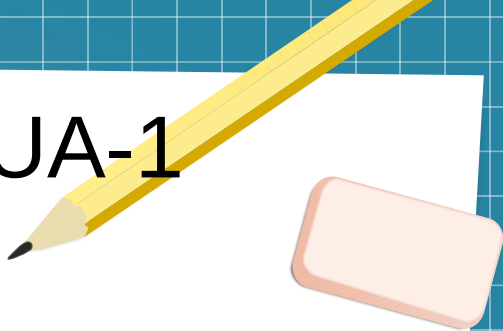
PDF 1.7 – PDF/UA-1

- 2008 standard
- Supported by most accessibility checkers
- Expects alt tags for math formulas
- Weakly supported by LaTeX

PDF 2.0 – PDF/UA-2

- 2020 standard
- Supported by few accessibility checkers
- Embeds MathML for math formulas
- Well supported by LaTeX
- Not supported by Ally
 - “6 month timeline” as of 4/8
 - “6+ month timeline” as of 9/15

Getting Started: PDF 1.7 – PDF/UA-1

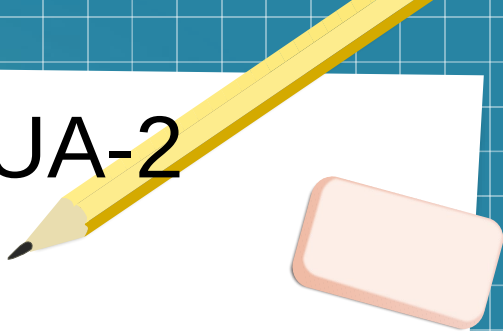


- Make sure your LaTeX installation is up to date
- Add code at the *beginning* of your TeX document.

```
\DocumentMetadata{  
  lang          = en,  
  Pdfversion    = 1.7,  
  pdfstandard   = ua-1,  
  tagging=on,  
  tagging-setup={math/mathml/luamml/load=false}  
}
```

- Run with LuaLaTeX

Getting Started: PDF 2.0 – PDF/UA-2



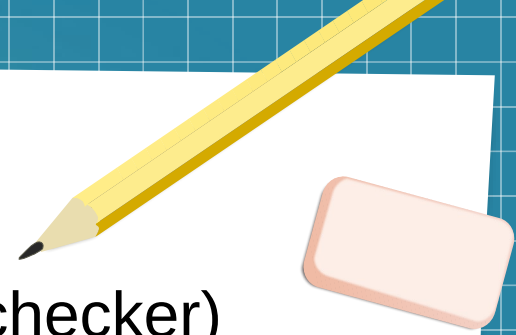
- Make sure your LaTeX installation is up to date
- Add code at the *beginning* of your TeX document.

```
\DocumentMetadata{  
  lang          = en,  
  pdfstandard   = ua-2,  
  tagging=on,  
  tagging-setup={math/setup=mathml-SE}  
}
```

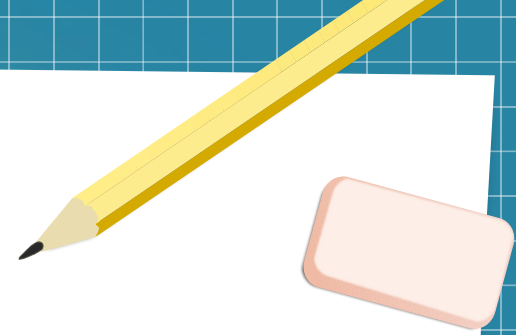
Run with LuaLaTeX

Checking Accessibility

- For PDF 1.7 – PDF/UA-1, use Ally (or your LMS checker)
- For PDF 2.0 – PDF/UA-2, use VeraPDF (linked)
- For either – open in Acrobat to examine tagging structure

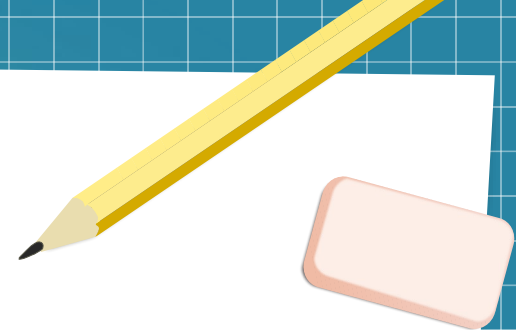


Alt Tags for Images



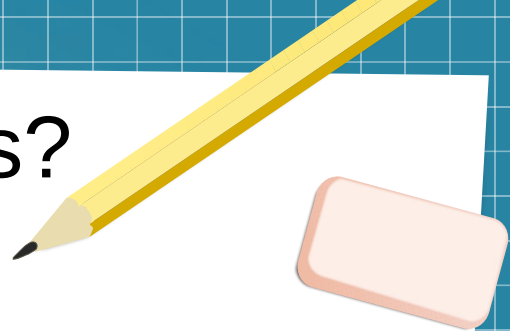
- The following commands support alt tagging:
 - `\includegraphics`
 - `\tikzpicture` (or the environment version)
- Include in the optional argument
- Alt text: `alt={alt text}`
- Decorative image: `artifact`
- Image of text: `actualtext=text`

Fun With Tables



- Accessible tables need headers.
 - `\tagpdfsetup{table/header-rows={1}}`
- Or, if they're not *really* tables, they should be marked as display.
 - `\tagpdfsetup{table/tagging=presentation}`
- In the preamble (applied to all tables) or before each table

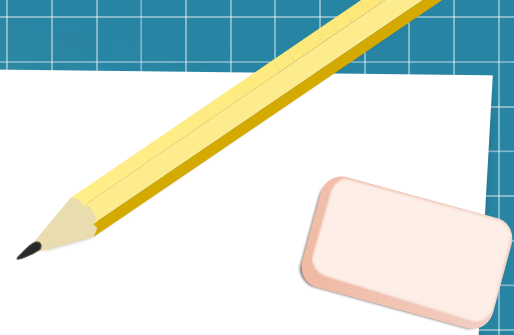
Do *You* Have 150 Worksheets?



- Use the command line! Run as a batch!
- Windows Powershell example (with TeX installation in \$Path)

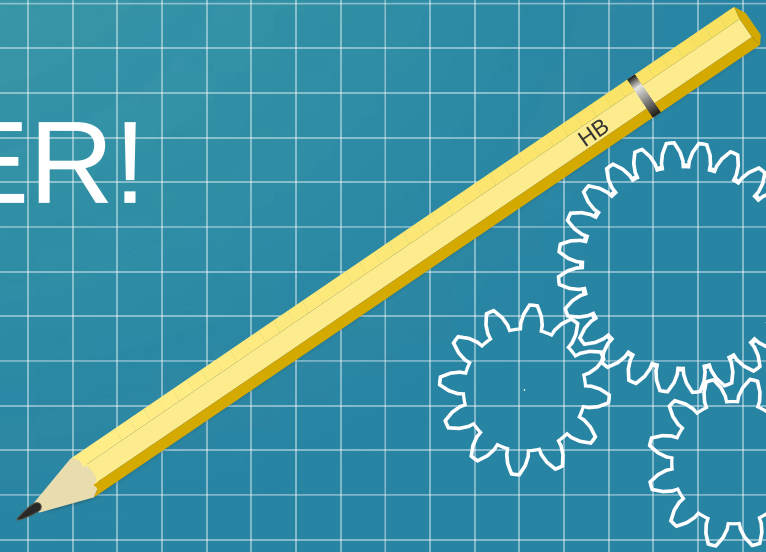
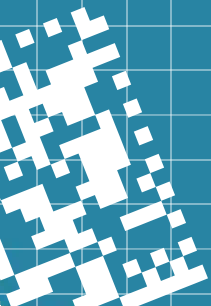
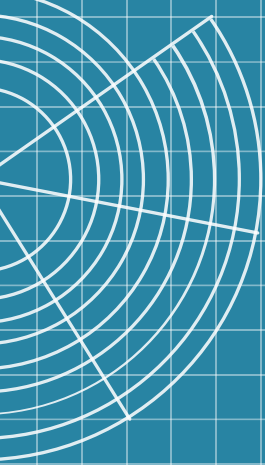
```
$files = Get-ChildItem -Path "." -Filter "In-Class*.tex"
foreach ($file in $files) {
    & lualatex.exe $($file.Name)
}
```

Going Forward

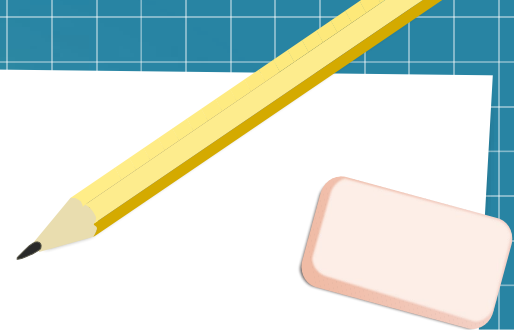


- LaTeX is a typesetting program.
 - Typesetting means **PRINT**
 - If it's not basically a print document, it shouldn't be in LaTeX
- Use PreTeXt for e-books
 - HTML is accessible
 - You can also get ebook formats (sweet!)
 - If you use TikZ, you kinda need to be on Linux

Oh No, My OER!

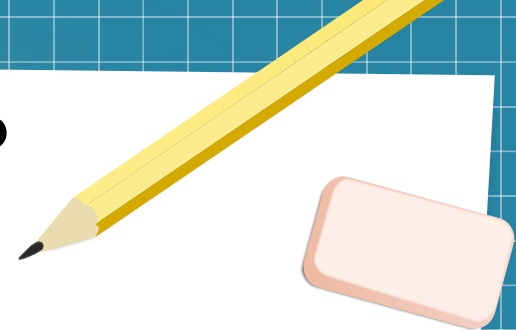


Don't Panic!



- It might not be as bad as you think.
- Run it through Acrobat's accessibility checker and auto-fix.
 - Be careful! Acrobat occasionally eats things
- If you have Word files, use the Word tricks.
- If you've *adopted* OER, check for an updated version.
- Done is good – aim for improvement, not perfection
- Collaboration! You're not alone.

Online Homework for OER?

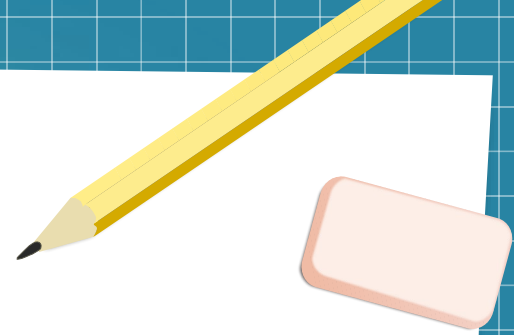


- MyOpenMath is good on accessibility.
 - Alt text your images.
 - Don't commit evil in your code.
 - Write accessible HTML.
- I made OER homework in my LMS!
 - Why? No, really, why?
 - HTML best practices will still help you, though.
- Alt text is easier here
 - Give enough information to solve the problem!

Links and Code Samples

- Get my links and sample files on my github

`https://github.com/amasaros/pdftalk2026/`



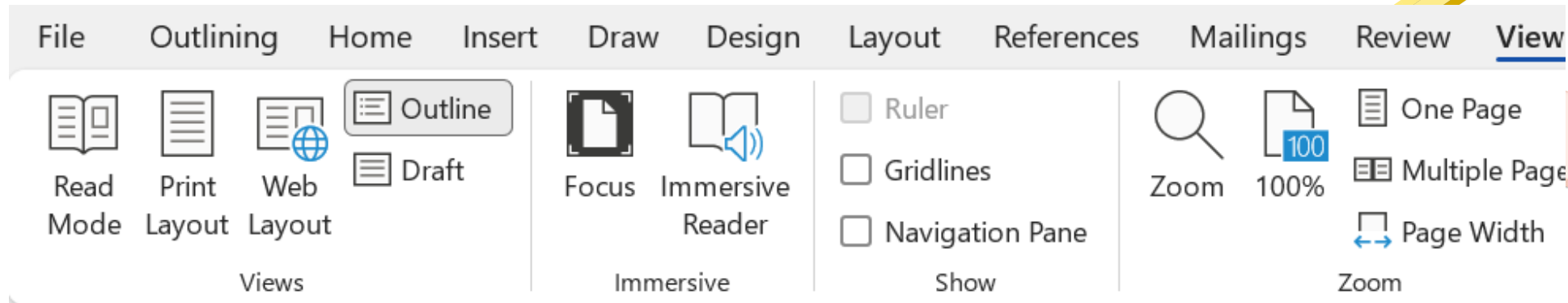


That's All, Folks!

America Masaros
Harper College

Email: `amasaros@harpercollege.edu`

Github: `https://github.com/amasaros/pdftalk2026/`



+ Section 3.3: Formal Logic

- + Concept 1 – Truth Values**
- + Concept 2 – Simple and Compound Statements**
- + Concept 3 – Symbolizing Statements**
- + Concept 4 – Truth Tables**
- + Concept 5 – Negation of a Statement**
- + Concept 6 – Conjunction**
- + Concept 7 – Disjunction**
- + Concept 8 – Conditional and Biconditional**
- + Concept 9 – Truth Tables to Symbolize Statements**

File Home Insert Draw Design Layout References Mailings Review View Developer Help Acrobat **Table Design** Table Layout

- ☒ Header Row ☐ First Column
☒ Total Row ☐ Last Column
☒ Banded Rows ☐ Banded Columns

Table Style Options

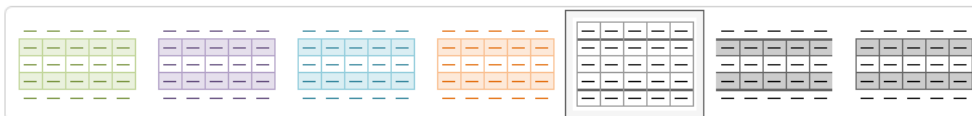
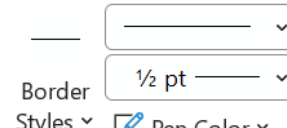


Table Styles



Shading



Border Styles

1/2 pt

Pen Color

Borders

102	108	81	96	103	91	90	96	86	92
84	92	90	103	115	93	85	116	87	106
85	88	106	104	102	98	116	107	102	89

Frequency and Relative Frequency for IQ Scores

IQ	Frequency	Relative Frequency
63.5 – 70.5	2	0.04 or 4%
70.5 – 77.5	2	0.04 or 4%
77.5 – 84.5	5	0.10 or 10%
84.5 – 91.5	9	0.18 or 18%
91.5 – 98.5	11	0.22 or 22%
98.5 – 105.5	9	0.18 or 18%
105.5 – 112.5	6	0.12 or 12%
112.5 – 119.5	4	0.08 or 8%
119.5 – 126.5	1	0.02 or 2%
126.5 – 133.5	1	0.02 or 2%
Total	50	1.00 or 100%

Search the menus

Cut

Copy

Paste Options:

Edit Picture

Change Picture

>

Group

>

Bring to Front

>

Send to Back

>

Link

>

Save as Picture...

Insert Caption...

Wrap Text

>

View Alt Text...

Size and Position...

Format Picture...

Layout

Position

Text Wrapping

Size

Wrapping style

In line with text

Square

Tight

Through

Top and bottom

Behind text

In front of text

Wrap text

Both sides

Left only

Right only

Largest only

Distance from text

Top

0"

Left

0.13"

Bottom

0"

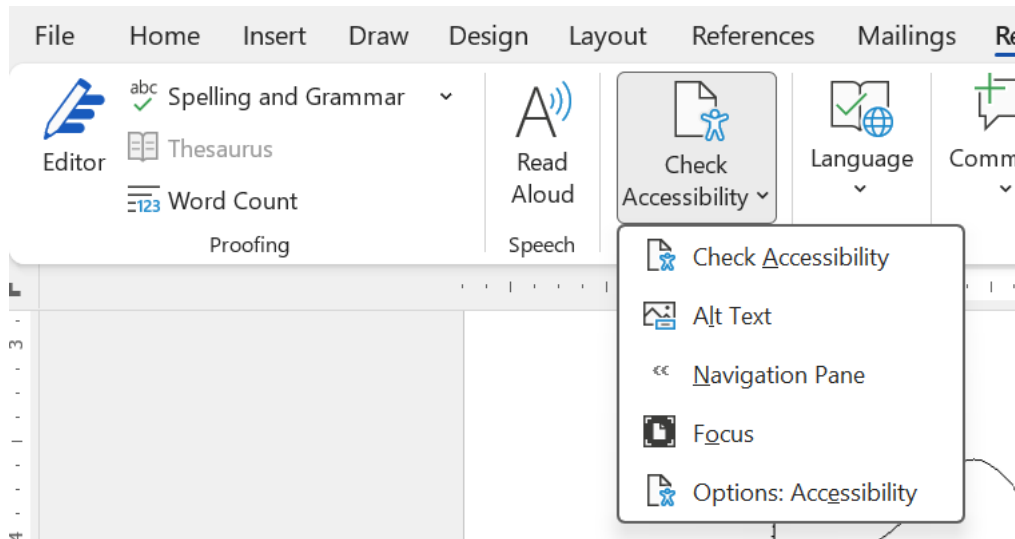
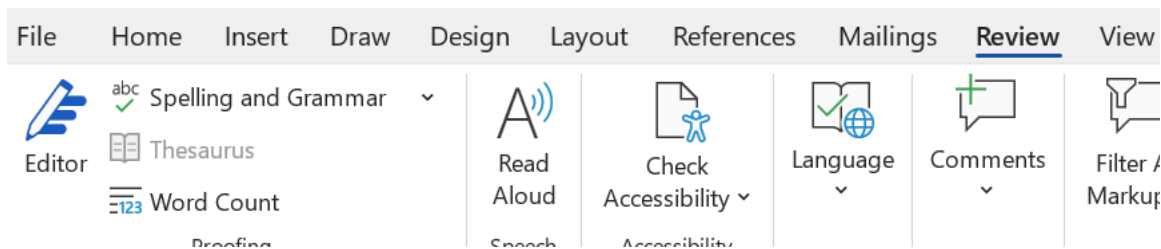
Right

0.13"

OK

Cancel

A yellow pencil with a black eraser and a pink eraser are positioned in the top right corner of the page. The pencil is angled diagonally, and the eraser is placed next to it.



Accessibility Assistant

Color and Contrast

Hard-to-read text contrast

Media and Illustrations

Missing alt text

Tables

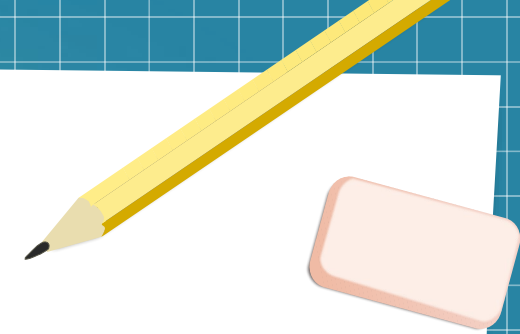
Missing table header

Use of merged or split cells

Document Structure

No headings in document

Unclear hyperlink text



Home

New

Open

Info

Save

Save As

Save as Adobe
PDF

Info

Section 1.8 - Normal Distribution

F: » Document Samples



Upload



Share



Copy path



Copy local path



Open file location



Protect
Document ▾

Protect Document

Control what types of changes people can make to this document.



Check for
Issues ▾

Inspect Document

Before publishing this file, be aware that it contains:

- Document properties, author's name, related dates and cropped out image data
- Headers and footers

Properties ▾

Size	463KB
Pages	12
Words	2157
Total Editing Time	3228 Minutes
Title	Section 1.8 - Normal Distri...
Tags	Add a tag
Comments	Add comments